

DP Manager User Guide (4th Edition)

Nov. 30, 2020

For questions or comments, contact the support team for the product under which this component is invoked.

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Preface

About This Document

The DP Manager User Guide gives you a brief overview of CDPL, the design of the CDPL GUI and its major interface elements. It shows you how to use the DP Manager to monitor and control a distributed process run.

Scope

The document covers everything you need to know about using the DP Manager. It is however not an authority on CDPL or distributed programming as such. Refer to Related Publications for information about CDPL.

Conventions and Acronyms

The following conventions and acronyms are used:

<i>Convention</i>	<i>Description</i>
[]	Denotes optional parameters
	Denotes choice between parameters
Control-c	Indicates a keyboard combination, such as holding down the Control key and pressing c
Edit > copy	Indicates a path to a menu command, such as opening the edit menu and choosing copy

<i>Acronym</i>	<i>Description</i>
CDPL	Common Distributed Processing Library
GUI	Graphical User Interface
SGE	Sun Grid Engine
LSF	Load Sharing Facility
RSH	Remote Shell
DP	Distributed Processing
SSH	Secure Shell
SH	Shell
RTDA	Runtime Design Automation
PBS	Portable Batch System
SLURM	Simple Linux Utility for Resource Management

Audience

This primary audience for this manual includes users who are new to Synopsys DP Manager.

Related Publications

For additional information about the DP Manager and CDPL, see

- DP Manager User Guide: Frequently Asked Questions
- CDPL User's Manual¹

¹ Currently only available to Synopsys R&D and field application engineers.

Introduction to CDPL

Common Distributed Processing Library (CDPL) is a ready-to-use library that is integrated with Synopsys EDA products and used to develop parallel and distributed processing workflows.

CDPL DP Building Blocks

CDPL assumes that an application process or processes have been partitioned into smaller tasks for distributed processing. This requires the application to divide a single process or a group of processes into smaller tasks that can be run concurrently. There may also be a final reconciliation and synchronizing task, as in the scatter-gather processing model. Once the application has been partitioned, the application is considered CDPL-Ready.

In the CDPL framework, the smaller partitions (i.e., tasks) of an application (e.g., EDA) process are collectively referred to as a **JOB**. The application program that creates, submits, and starts the JOB is called the **MASTER**. The application program that accepts and processes the smaller tasks is called the **WORKER**. MASTER defines tasks and any task dependences using CDPL API, and dispatches tasks to appropriate WORKERS through CDPL. CDPL provides IPC mechanisms and API for handling all required communication between MASTERS and WORKERS.

The MASTER and WORKER programs' application-specific logics as well as JOB contents and task scheduling constraints are in the application domain, and are controlled and managed by the application teams.

A CDPL-enabled parallel and distributed application may consist of a number of executables, each of which is referred to as an ENTITY. The table below explains some details of entities that commonly work together in a CDPL-enabled application:

ENTITY	Description	Owner
MASTER	Creates a job and submits it. Typically, it also processes the results obtained from the server after the job is completed successfully	Application Team
WORKER	A basic entity that can be assigned a task. Typically, a worker is a program supplied by the application that is run on a selected list of hosts specified by the end-user.	Application Team

MONITOR	Monitors the status of submitted jobs by communicating with all relevant entities	CDPL Team
THINWORKER	A worker that launches a third-party binary supplied by the application	CDPL Team

Table 1 : CDPL Entities

CDPL Monitor

The CDPL Monitor Entity provides an API to monitor the status of submitted jobs. This API provides the following functionality:

- Monitoring of submitted jobs
- Retrieval of job and task status information
- Retrieval of worker information like memory, CPU utilization on demand.
- Execution Control –Configuration, Start, Stop, Logging etc. (common to all CDPL entities).

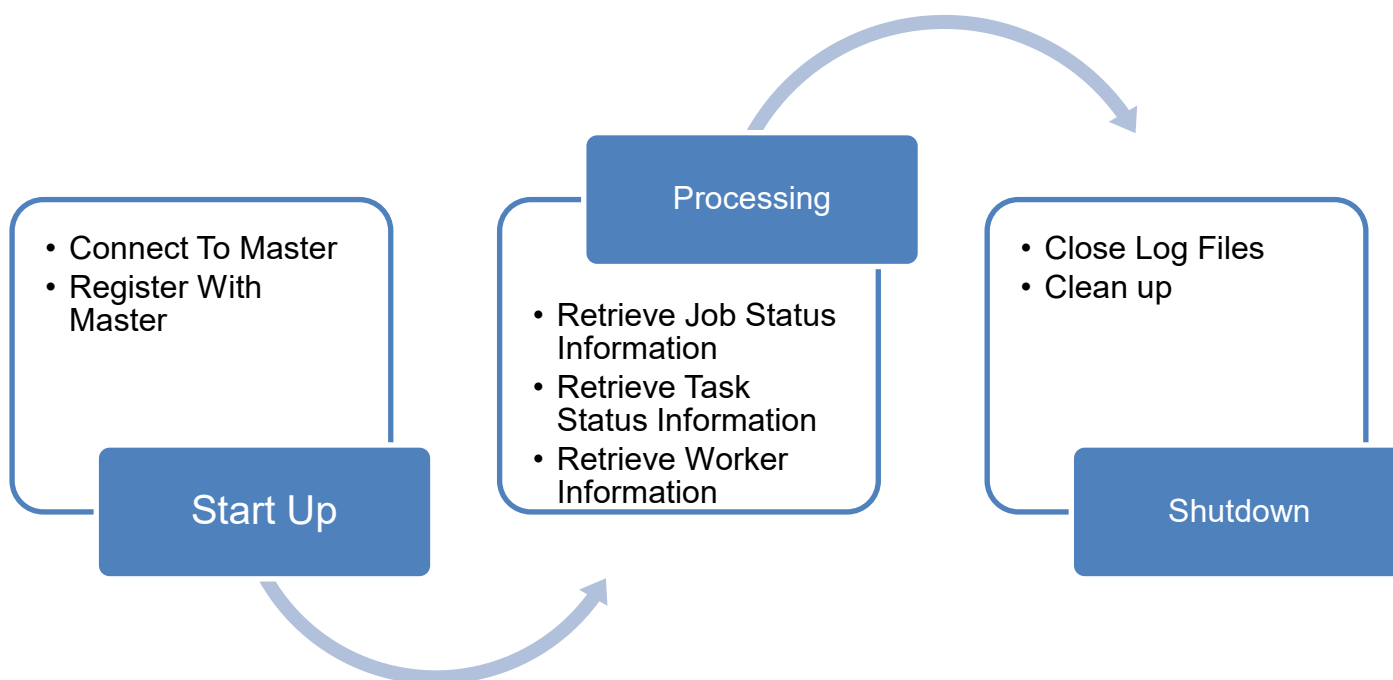


Figure 1: Monitor Execution

The DP Manager GUI

The DP MANAGER monitors all activities performed by the CDPL and provides limited interactive operations to control a few other events. The DP Manager interface has the following high-level modules.

Serial No.	Interface Module	Brief Description
1	DP Monitor	Monitors a DP Job for Task status information. Provides an interface for starting and killing a DP run.
2	Host Status	Display real time worker system metrics such as memory and CPU usage graphically.
3	DP Analyzer	Analyze a DP run to collate worker/task statistics and measure efficiency its efficiency.
4	DP Log Browser	View DP log files.
5	Configuration Editor	Create and edit DP hosts files and DP worker files
6	Cluster Manager	Create and manage sets of DP hosts
7	Help Assistant	View Help documentation.

Table 2 : DP Manager Module List

DP Monitor

The DP Monitor monitors a DP run for job and task status information to keep the user updated about the state of the job, the progress of its tasks and other useful information. The DP Monitor is displayed by default when the DP Manager starts. If the window is closed, it can be reopened through the View->DP Monitor menu.

This module communicates with the DP Master to periodically retrieve and display job information. The DP Monitor collates this information and presents it visually in a tree structure. A typical job tree structure is shown below.

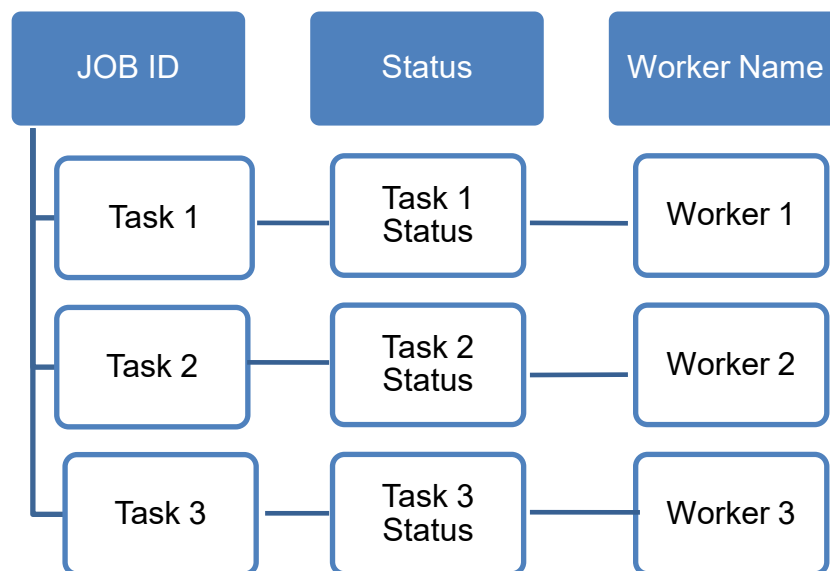


Figure 2: The Job Tree Structure

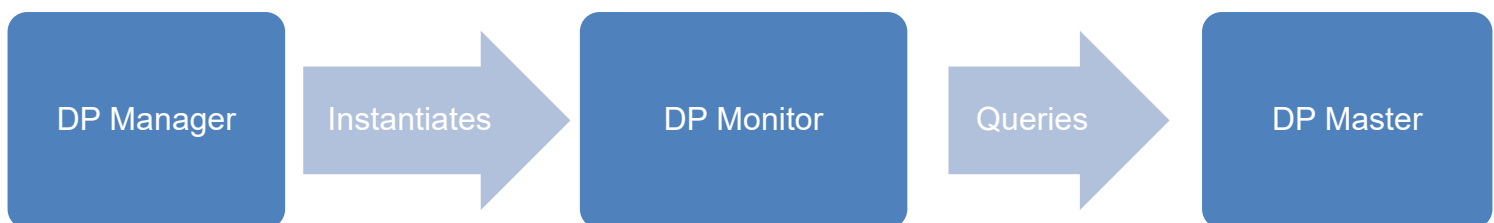


Figure 3 : DP Monitoring module process flow

Job/Task Status

The major states of a job or task are:

1. NEW
2. SCHEDULED
3. STARTED
4. INPROGRESS
5. RUNNING
6. COMPLETED
7. STOPPED (can be restarted)
8. KILLED (cannot be restarted)
9. INTERRUPTED (Run was interrupted externally, e.g. by Ctrl-C)

The DP Monitor uses a simple color coding mechanism to indicate the status of a DP Job/Task.

Job/Task Status	Color
SCHEDULED/INPROGRESS/NEW	Blue
COMPLETED	Green
KILLED/STOPPED	Red
Others	Black

Table 3 : DP Monitor Color Coding Mechanism

Connecting to a DP Run

The DP Monitor connects to a running job by searching for a broadcast file in the default location **\$HOME/.synopsys/cdpl**. This file is created by the DP Master. It contains job information and parameters that the DP Monitor uses to communicate with the Master.

DP Manager GUI searches only the broadcast files that were generated in the last 30 days. This feature was added to resolve the issue (Case Number: 9001200112), where the DP manager GUI failed to load as a result of a huge number of broadcast files present in the **\$HOME/.synopsys/cdpl** directory.

The **CDPL_BCASTDIR** environment variable can be used to set a different location for the broadcast files.

Note: When **CDPL_BCASTDIR** is used, it is essential that both the DP Manager and Master use the same setting for **CDPL_BCASTDIR**.

Monitoring Jobs

The DP Manager provides read access to all DP activities. This information is encapsulated in the “DP Monitor” window. This includes

- Status of running jobs and tasks.
- Details about the DP workers (ID, PID and host).
- Job summary
- An interface to the DP master and worker logs.
- CPU and memory status of the host machines.
- The setting of the DP Info Directory. The DP Info Directory is the location of the DP broadcast files. This is \$HOME/.synopsys/cdpl by default, or set via CDPL_BCASTDIR.

The DP Manager monitors a DP Run for job and task status information and keeps the user updated about the state of the DP Job, the progress of its tasks and other useful information. The DP Monitor window is shown below.

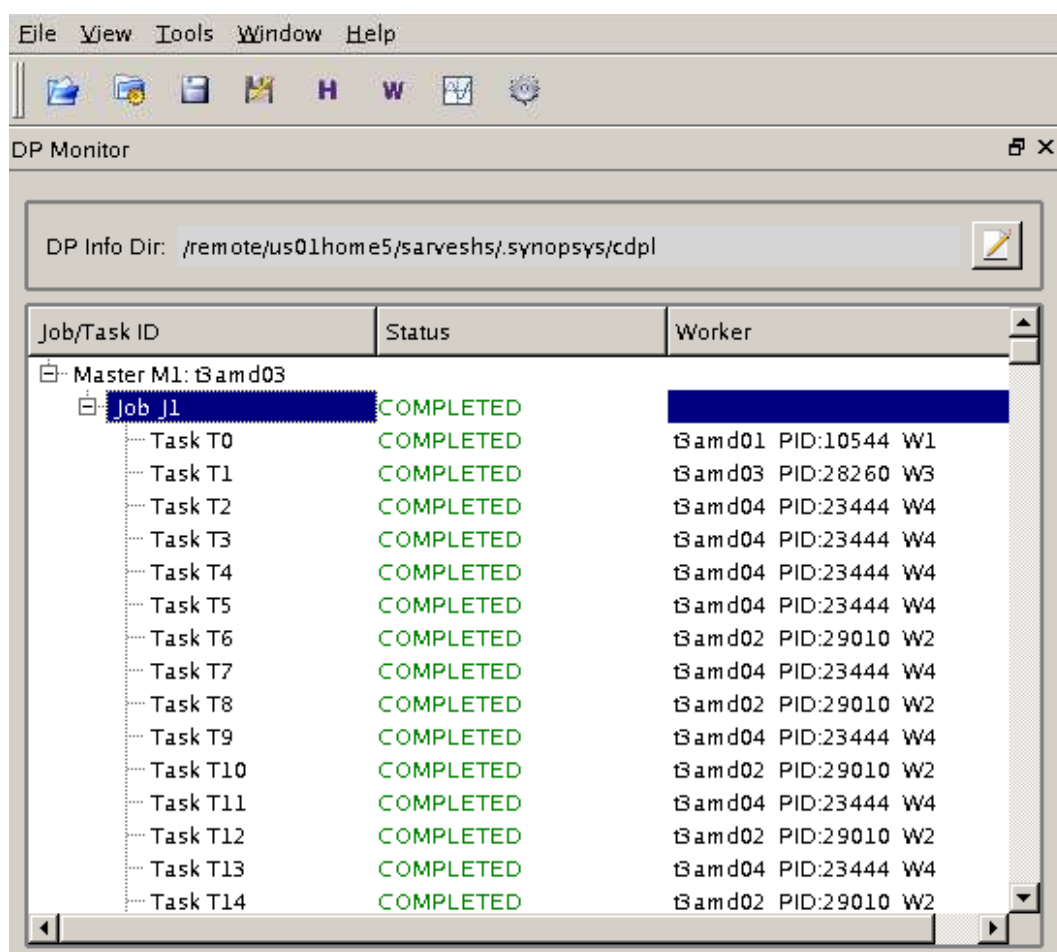


Figure 4

The DP Monitor also displays details about the individual job in the Job Summary section shown below. The Job Summary displays the name of the job currently summarized, and shows its overall progress in a progress bar.

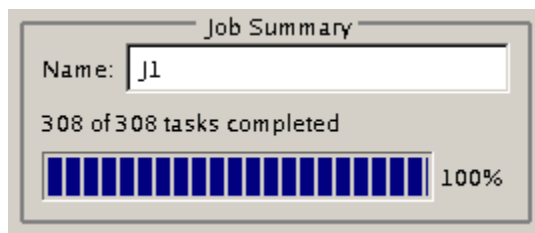


Figure 5

The DP Monitor could be connected to several DP runs, or a single DP run may contain multiple jobs. To view the summary of a particular Job, simply click on its name in the tree as shown below. In this figure, the DP Monitor is connected to two DP runs. Each DP Master is a top-level item in the job tree. By default, the Job Summary window shows the status of the most current Job. Click on the name of the job from the drop down tree to view its summary.

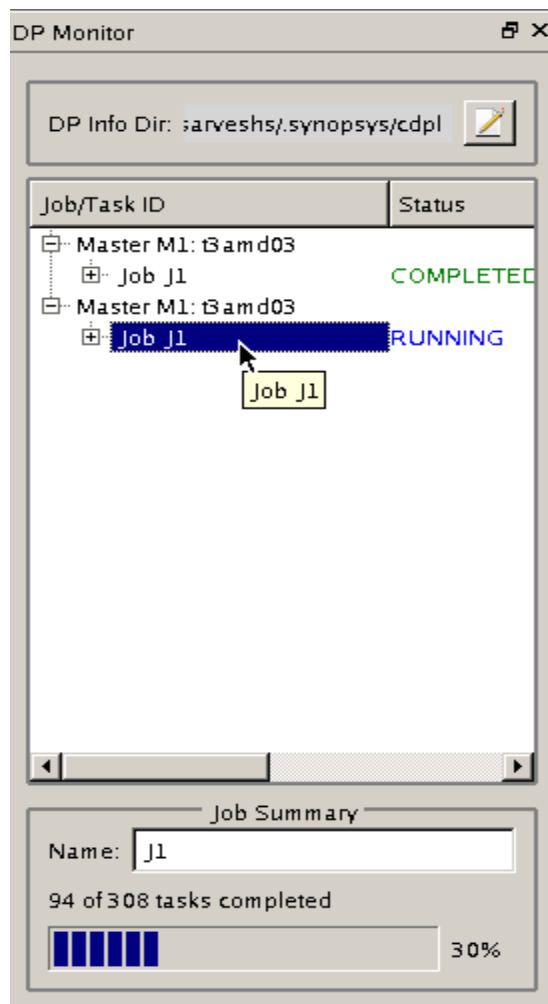


Figure 6 : Multiple DP Runs

Stopping, Killing and Restarting Jobs

The DP Monitor provides a context menu to

1. Stop a running Job
2. Kill a running job
3. Start a stopped job

To access the context menu, right click on the name of the job in the Job Tree displayed in DP Monitor window.

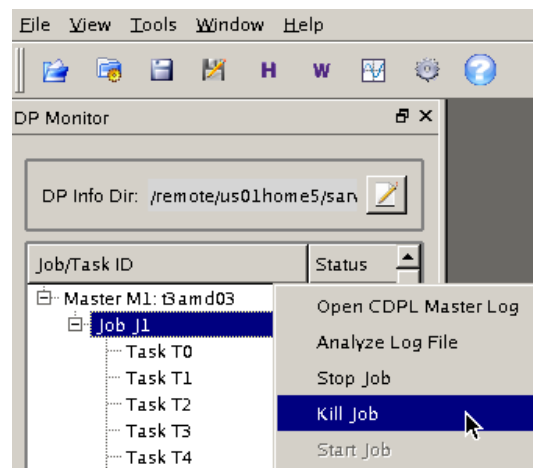


Figure 7 : Killing a Job

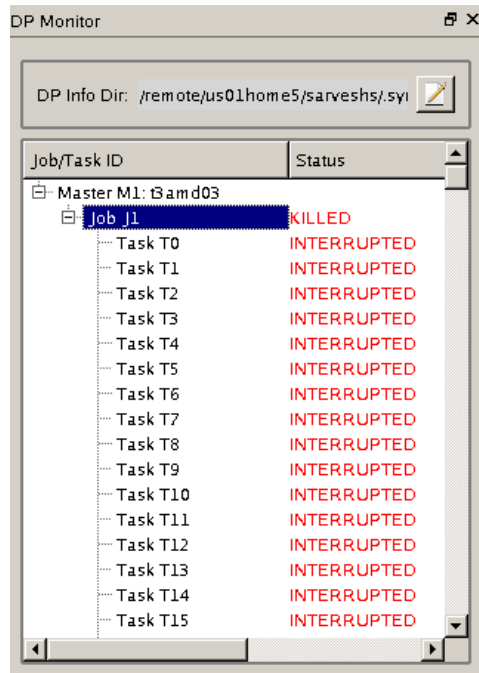


Figure 8 : Killed Job Status

Note: The application needs to explicitly handle the STOP event initiated by DP Monitor. Check with your application support team whether this feature is supported.

Starting a DP Run

You start a DP Run from the DP Manager in the following two ways.

1. Use the `--run` command line option to pass the executable and its arguments to the DP Manager binary:
 Example: `dpmanager --run dpprogram SAMPLE_INPUT_1 SAMPLE_INPUT_2`
 You can pass a startup script to the DP Manager instead.
 Example: `dpmanager --run START_DEMO.sh`
2. Go to Tools->Start DP Run to select a startup script for execution. You can also use the the Start DP Run tool button. These are shown below.

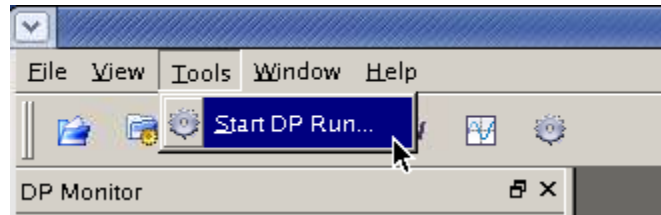


Figure 9

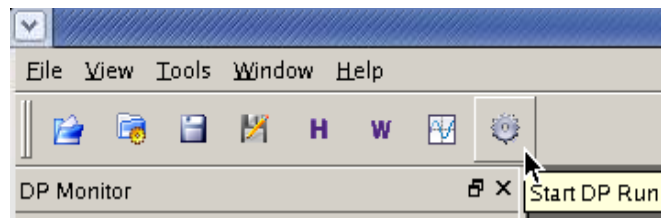


Figure 10

Clicking on either the “Start DP Run” menu item or the tool button, will display a file chooser. Select the startup script from the file chooser and press Open to start your run.

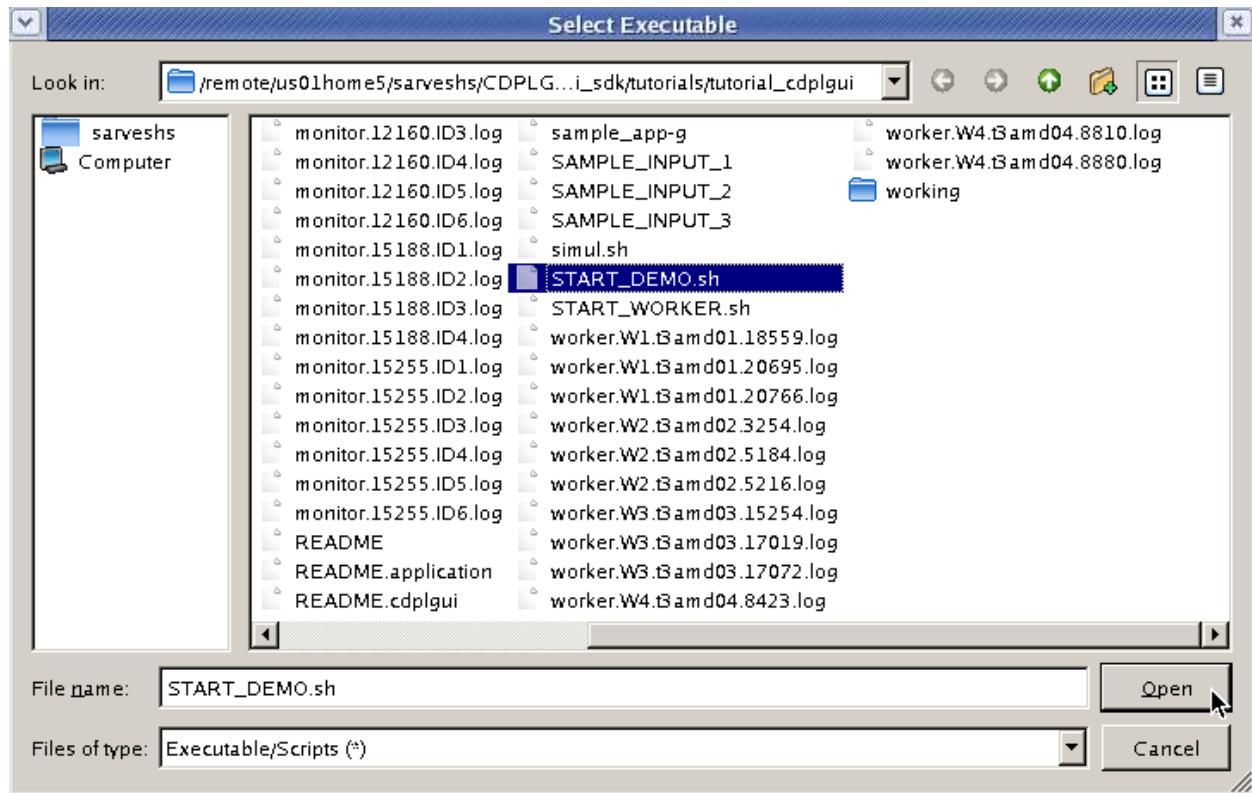


Figure 11 : Select Executable for a DP Run

Standard output and standard error messages can only be seen if you start a DP Run using the DP Manager. To view the output, Go to View->DP Process and select the name of your startup script from the sub menu as shown below:

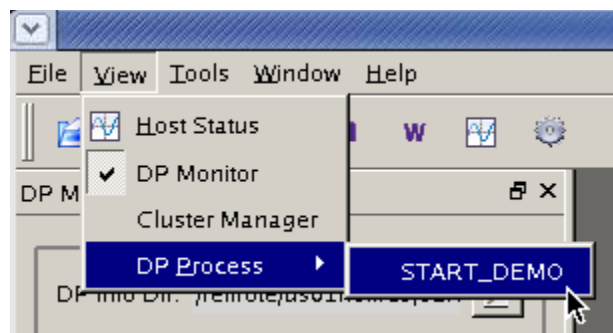


Figure 12

Clicking on the startup script name (START_DEMO in this example) will display its output window (shown below). The Output window displays the name of the executable, its PID, the full path to the executable and its standard output and error messages.

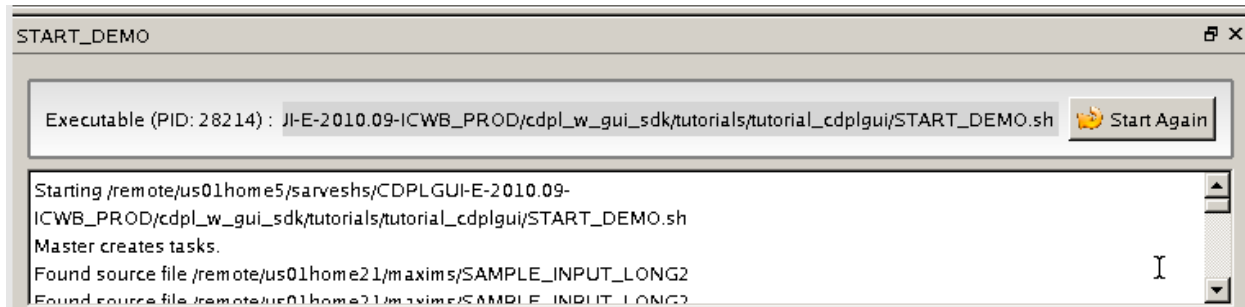


Figure 13 : DP Process Output

You can use the “Start Again” button to restart your DP run without having to search and select your startup script again. It is activated at the end of the run.

Host Status

The Host Status module displays relevant information about the hosts used for the DP Run. Information such as the CPU/Memory utilization, number of workers, current load, etc. are encapsulated in the “Host Status” Window.

To View the Host Status window, go to View->Host Status or click on the Host Status icon in the tool bar.

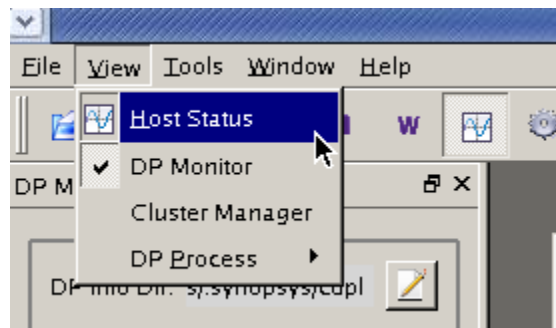


Figure 14

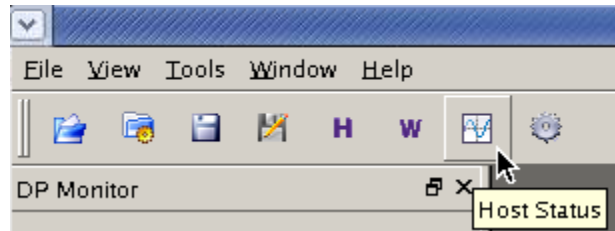


Figure 15

Use the DP Monitor to control whether the Host Status window displays aggregate information about all the hosts in the DP run or information about a single host.

When the job is selected in DP Monitor, the Host Status displays information about all the hosts.

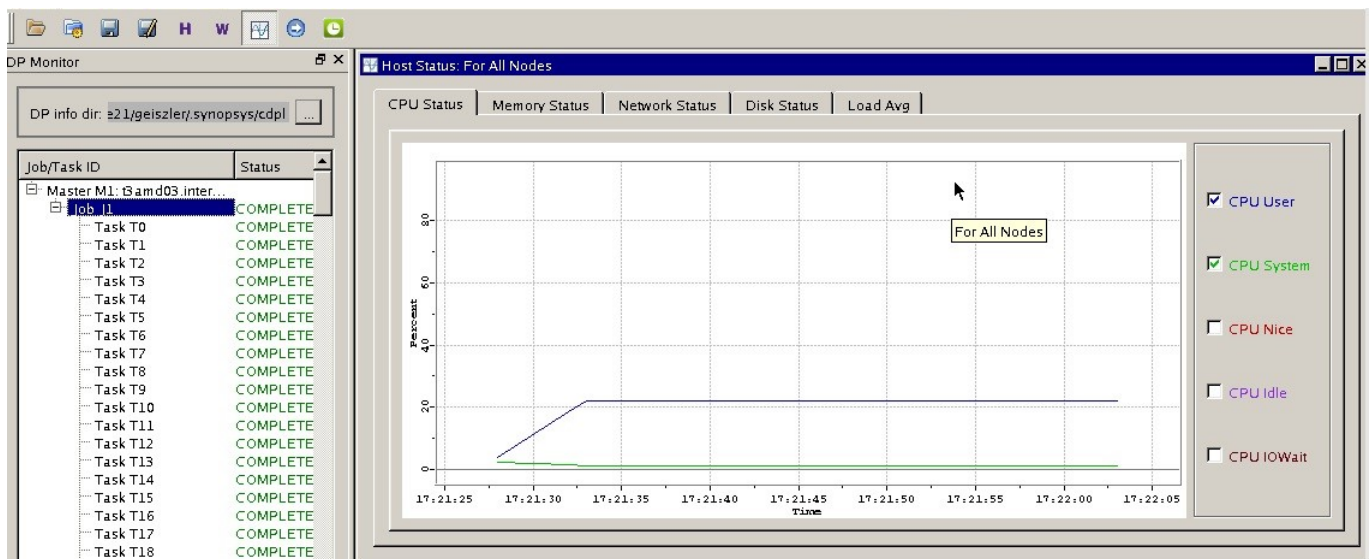


Figure 16

When any name on a task row is selected, the Host Status displays information about the host system of that task and its worker.

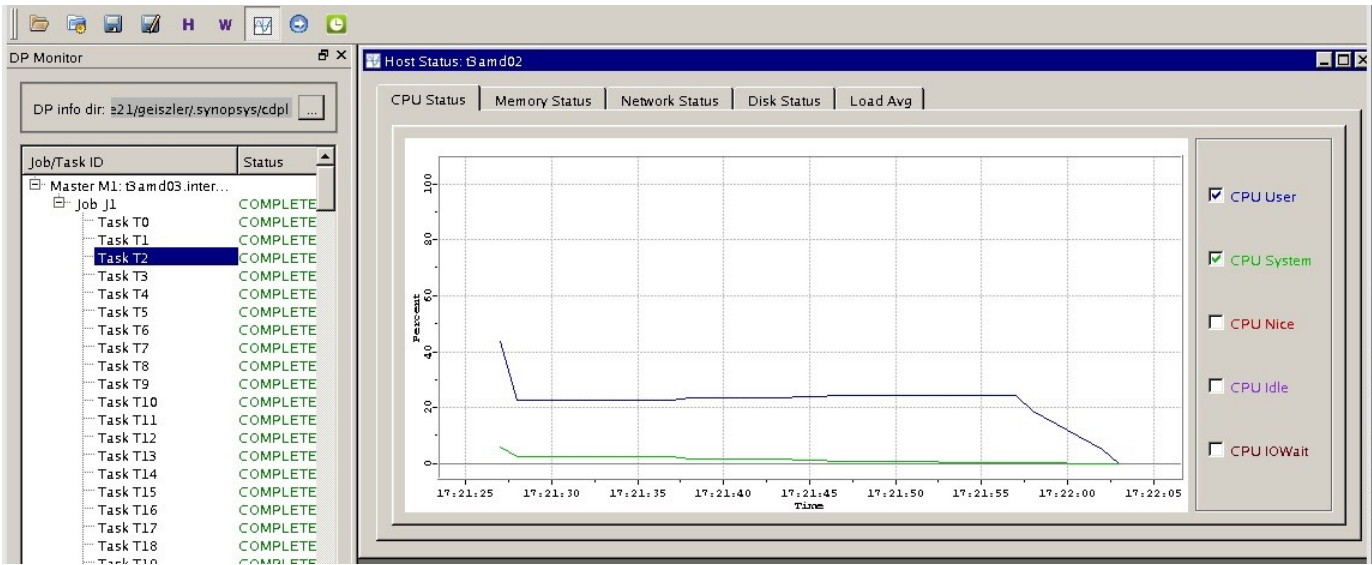


Figure 17

The Host Status window displays the CPU, Memory Utilization, Network Status, Disk Status and Load Average in separate tabs. It also shows miscellaneous information about the host such as its current load, current available memory, number of workers in the host, etc. as a tool tip. The Host Status window is shown below.

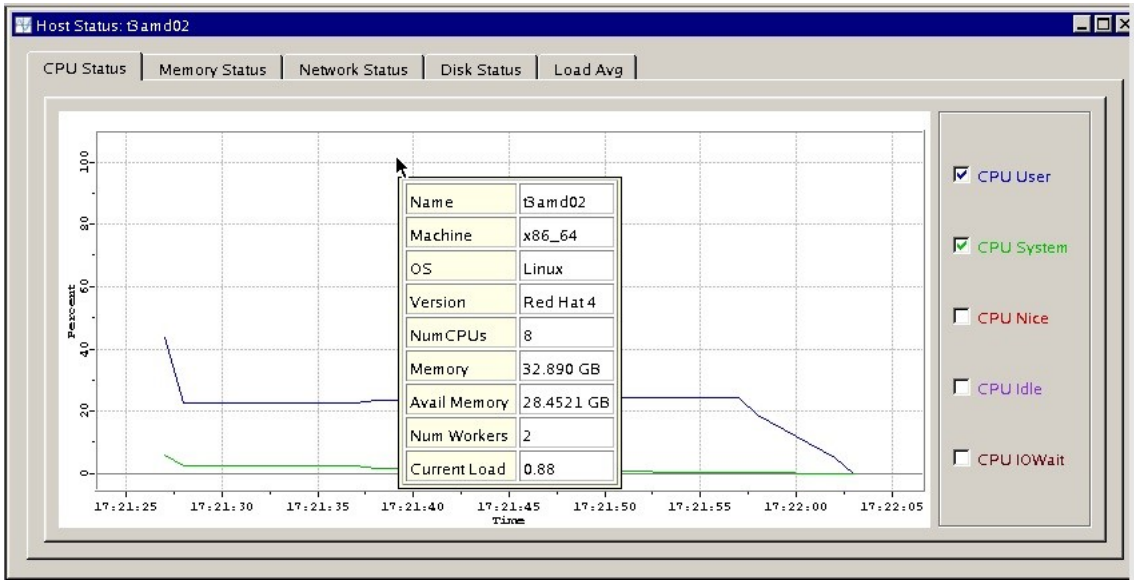


Figure 18

The metrics represented in the CPU/Memory Usage Vs Time curve graphs are:

- CPU Metrics in Jiffies (1/100 of a second for x86 systems)
 - System : processes executing in kernel mode
 - User : normal processes executing in user mode
 - Nice : normal processes executing in user mode with low priority
 - Wait : waiting for I/O to complete
 - Idle: Idle Tasks
- Memory in KB
 - Memory Used
 - Swap Used

Benefits

- Visually represents system statistics of all worker systems or individual worker systems.
- Provides information for analysis of a DP Run. This helps users understand the amount of memory and CPU their tasks are utilizing. They may use this information to redistribute their tasks more efficiently.

DP Analyzer

One of the primary motivations behind developing a DP Scheme is to increase the efficiency of EDA Applications. CDPL strives to meet this objective by:

- Efficient management of computer resources such as CPU's, Task (process) locations
- Ensuring Optimum Task to CPU mapping
- Allowing end users to dynamically change worker configuration
- Managing multiple jobs and job stages with the same resource environment.
- Minimizing data interchange congestion.
- Error detection and recovery.

Worker Efficiency

DP Analyzer is a module that outputs a measure of **total worker occupancy** as **efficiency metric**. In order to understand the advantages offered by DP, users would often like to compute this efficiency measure for their DP scheme.

For example, if the DP system was running for 100 seconds, and 10 workers ran an average of 95 seconds executing tasks, the overall efficiency will be 95%. This can be used to roughly estimate the amount of total time that workers were running, but not working on a task. The DP Manager uses DP Analyzer to display this information.

Analyzing a DP Run

The preferred way of analyzing a DP run in the current session of the DP Manager is to right click on the name of the job and select the “Analyze Log File” option from the context menu. This is shown below.

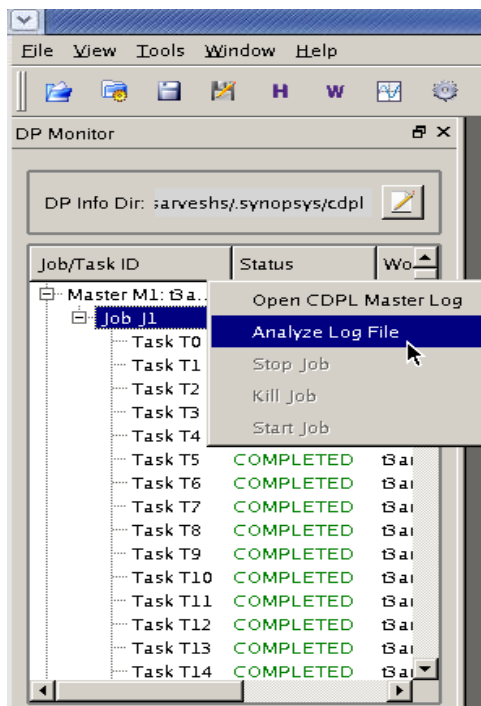


Figure 19

$$\text{Efficiency} = 100 - [(\text{master running time} - \text{average worker time} / \text{master running time}) * 100]$$

Or

$$\text{Efficiency} = \text{average worker time} / \text{master running time} * 100$$

You can analyze a log file not visible to the current session of DP Manager (a log file from a past run perhaps) by using the file menu File->Open and Analyze (Figure 20) or by using the Open and Analyze icon in the tool bar as shown below (Figure 21).

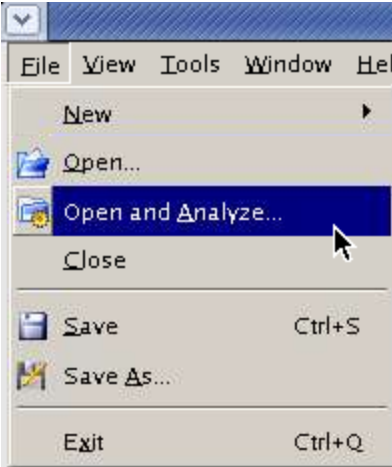


Figure 20

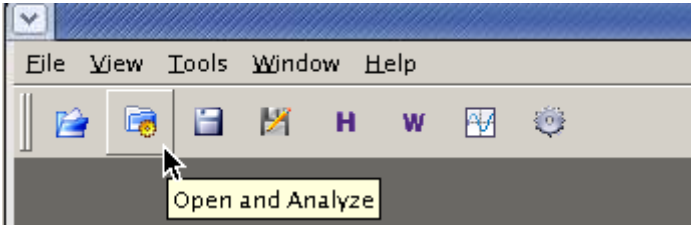


Figure 21

Analyzer Views

The DP Analyzer can display the output in a Gantt chart, a Tree Map or a Summary view. The different views of the output are shown below.

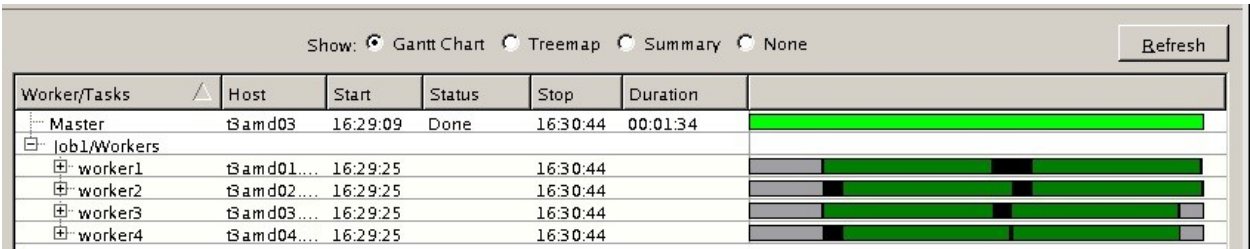


Figure 22

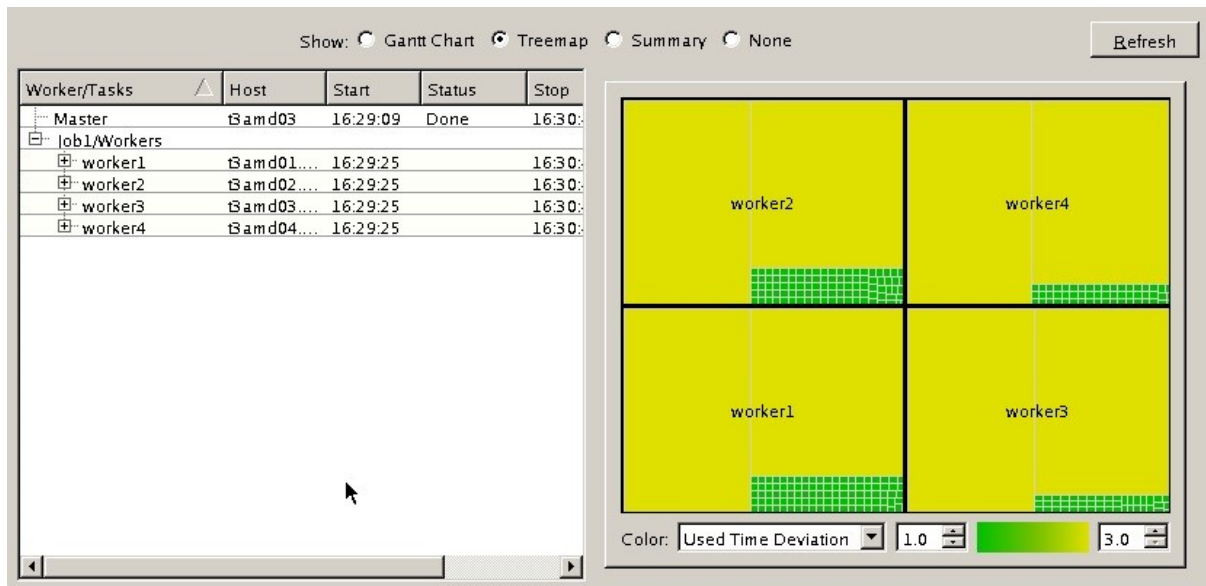


Figure 23

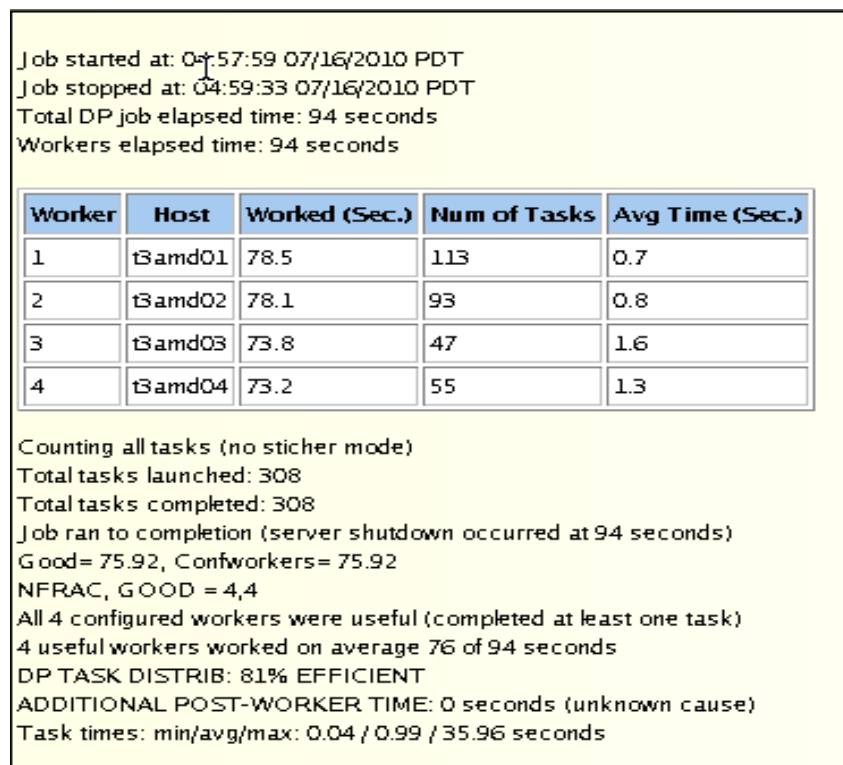


Figure 24

DP Log Browser

The Log Browser allows you to view DP log files and perform a keyword search. The DP log files contain a lot of information recorded from DP activities and isolating important events in the log files can be cumbersome. To make debugging easier, the DP Log Browser filters DP log files and highlights key events.

You need not search for DP log files. The DP Manager retains the location of the DP Master and Worker log files across runs started in the same session. You can view the logs while the DP process is running as well as after the run has completed.

Viewing a Master Log

To view a DP Master log file, right click on the name of the Master or on any of its jobs as shown below.

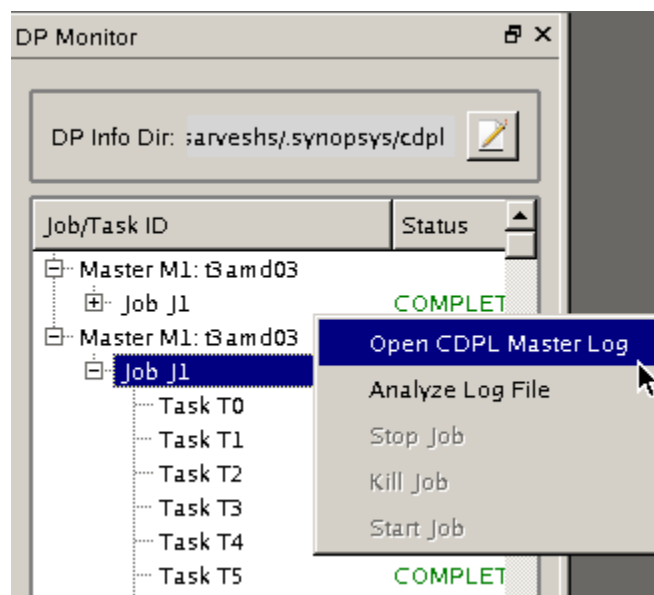


Figure 25

The log file will then be displayed in a new “DP Log Browser” window to the right. You can use the search box provided to isolate keywords.

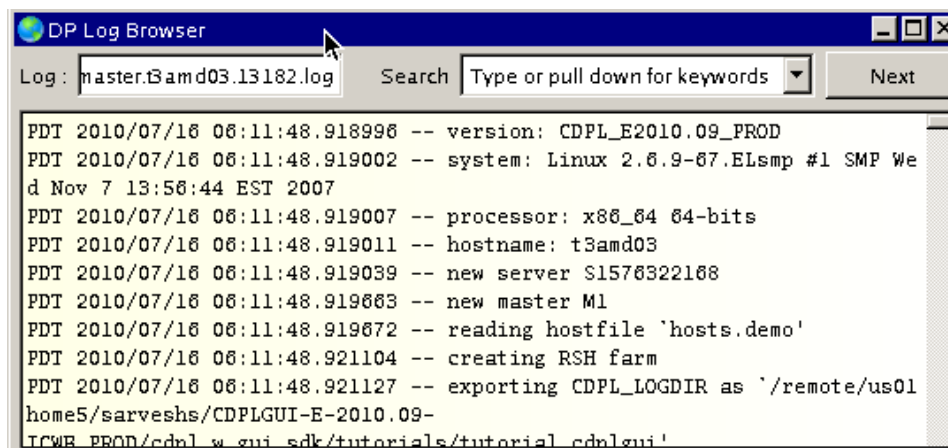


Figure 26

Viewing a Worker Log

To view a DP worker log file, right click on the name of the worker in the Worker column to open the worker log in a new DP Log Browser window as shown below.

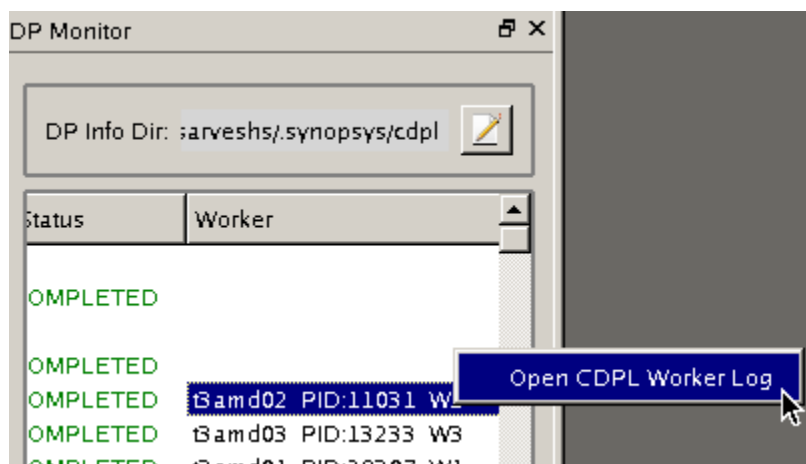


Figure 27

The DP log files are also accessible through the File->Open menu.

Note: For very short DP Runs (says 5 seconds) the DP Manager may not be able to retrieve all the information required to display the worker log in time. In such cases, only the host name is displayed in the worker column as shown in the figure below. The DP Manager did not get enough time to retrieve the worker ID and its PID. Hence in such cases, you may not be able to right click on the worker name to display its log.

Note: The DP Manager cannot display worker information for the tasks that have already completed when the monitor connects to the job master. The information is not available. The worker column will show “Not available” for these tasks.

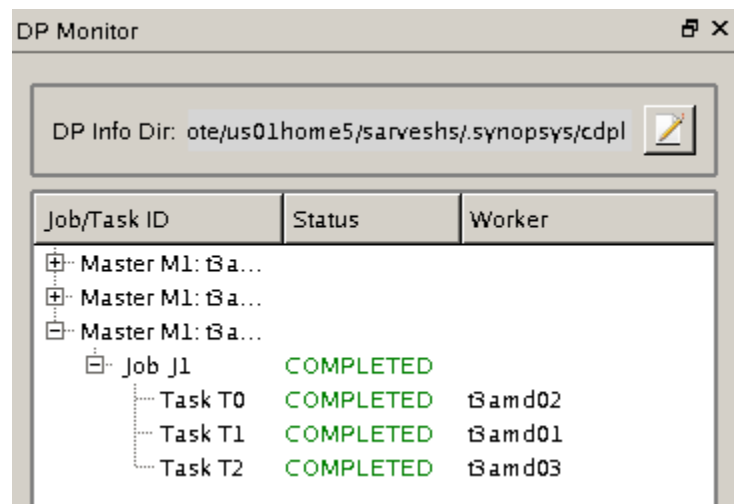


Figure 28

DP History

The DP Manager provides the DP History menu for viewing and managing DP Jobs. Through the DP History menu, you can view, manage and re-run previously executed DP jobs.

To view the DP History, click on View->DP History. The DP History window will come up.

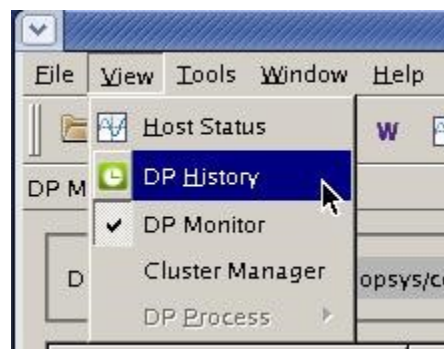


Figure 29

The DP History window is organized by Job History and Command History.

Job History

The Job History shows the list of recently run DP jobs. This includes jobs started from DP Manager and jobs started in the background. It allows you to select a job and Analyze or Delete it through buttons at the bottom of the window. This is shown in the following figure.

	Date	Time	Master ID	Master PID	Status
1	27.10.2010	09.01.16	Master M1: t3amd03	21076	✓ done
2	27.10.2010	08.46.33	Master M1: t3amd03	711	✓ done
3	26.10.2010	09.32.35	Master M1: t3amd03	16208	✓ done
4	26.10.2010	09.31.53	Master M1: t3amd03	16072	✓ done
5	26.10.2010	09.27.56	Master M1: t3amd03	15095	✓ done
6	26.10.2010	09.26.47	Master M1: t3amd03	8592	✓ done
7	26.10.2010	09.23.37	Master M1: t3amd03	8063	✓ done
8	26.10.2010	09.19.31	Master M1: t3amd03	7413	✓ done
9	26.10.2010	09.16.07	Master M1: t3amd03	6850	✓ done
10	26.10.2010	09.12.51	Master M1: t3amd03	6372	✓ done
11	26.10.2010	09.09.40	Master M1: t3amd03	5906	✓ done
12	26.10.2010	09.06.28	Master M1: t3amd03	5403	✓ done
13	26.10.2010	09.05.46	Master M1: t3amd03	5237	✓ done
14	26.10.2010	09.02.31	Master M1: t3amd03	4697	✓ done
15	26.10.2010	09.02.04	Master M1: t3amd03	4625	✓ done

Figure 30

To Analyze a job, select the job and click on Analyze. This will bring up the DP Analyzer window for the job. To Delete a job, select the job and click on Delete. This will delete the DP log and broadcast files for the job.

The “Delete All” button will delete all the jobs in your job history at once. This includes all the DP log files associated with those jobs.

Command History

The Command History window shows the commands that were recently run from DP Manager Tools->Start DP Run menu. To re-run a command from the Command History, select the row of the command and click on the Run button at the bottom of the window. The Command History window is shown below.

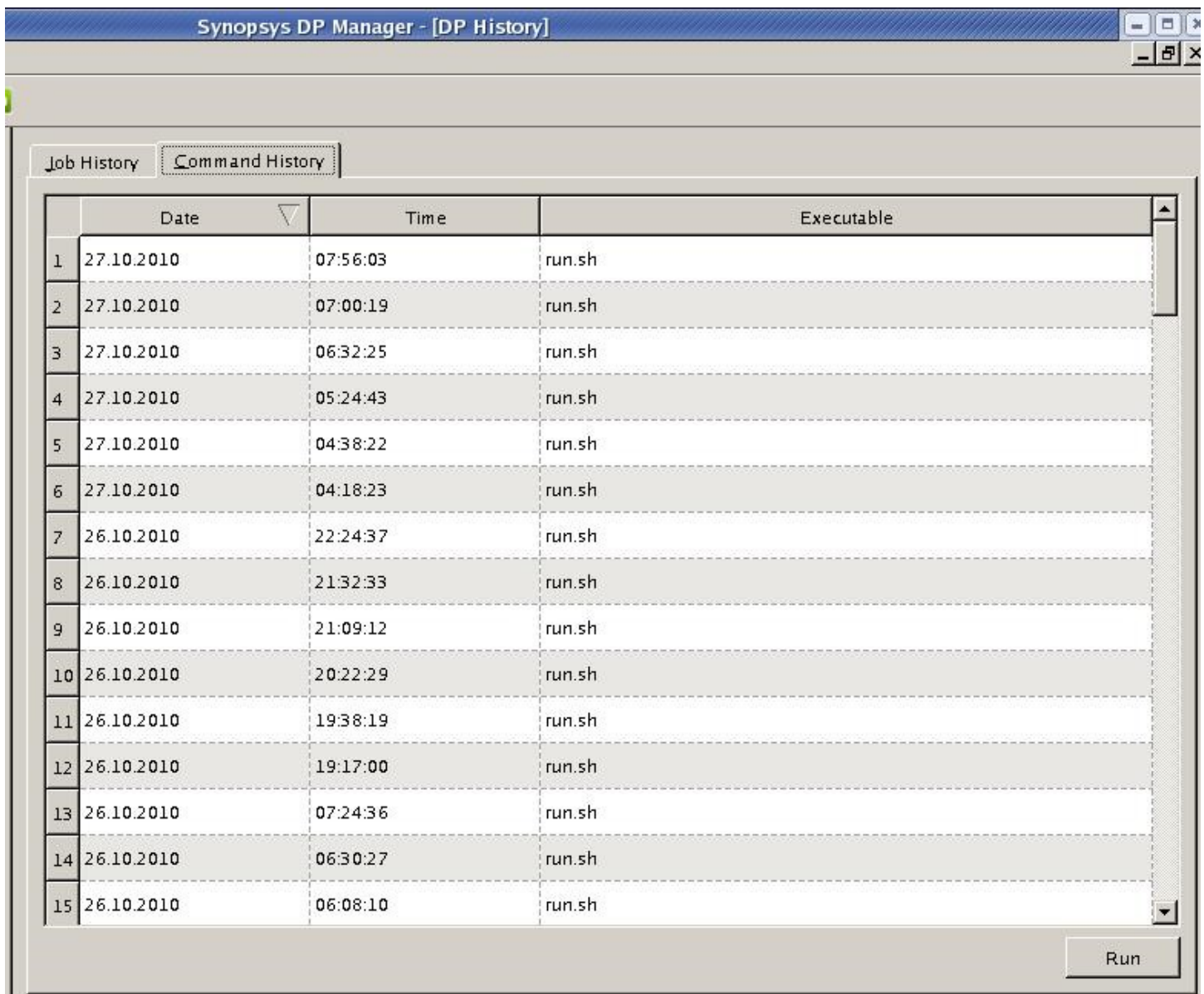


Figure 31

DP History Preferences

From Tools->Preferences, you can specify the number of jobs and commands that will be retained in the list. This is shown in the following figure.

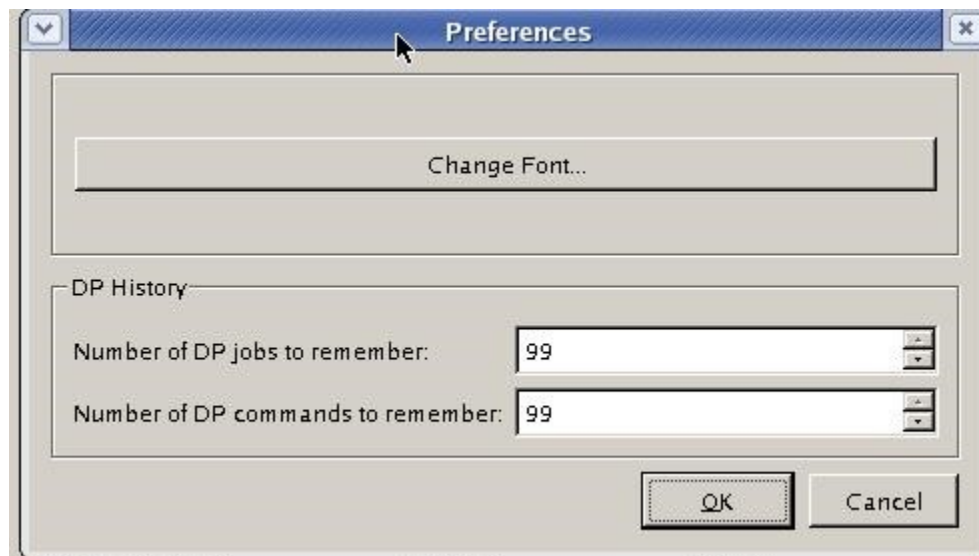


Figure 32

Configuration Editor

The DP Manager provides a Configuration Editor for creating and editing DP configuration files such as the hosts file and workers information file. The Configuration Editor is a convenience to relieve the user from the task of manually creating error-free configuration files. It provides an intuitive user interface to open, edit and save configuration files. It also provides some validity checks on the configuration data.

Hosts File

CDPL supports 7 types of transports that the Master can use to dispatch workers on multiple remote nodes: RSH (remote shell), SSH (secure shell), SH (shell), Platform-LSF (Platform Load Sharing Facility), SGE (Oracle Grid Engine), PBS (Portable Batch System) and RTDA (Runtime Design Automation). Note that Oracle Grid Engine was formerly known as Sun Grid Engine.

Depending on the system chosen by the user, CDPL expects different types of system-specific information. The Hosts information file is a consistent way to communicate information to the DP system for any of the above job dispatch mechanisms.

CDPL can handle only one job-dispatch mechanism per session. For e.g. if “RSH” is the preferred mechanism, the Hosts information file should contain a list of valid hosts. Alternatively, if SGE is the preferred mechanism, the Hosts information file contains the SGE command and options and the number of slots (processors) required for the parallel job to execute.

A typical hosts file has the following format:

```
# HOSTS INFORMATION FILE (RSH)
# Each valid host definition occupies an entire line.
# Line with '#' as the 1st char is a comment line.
# Format of each line:
# flag| hostname| slots| tmpDir| protocol | command
# For example:
1 | teralab-x15 | 4 | /remote/users/tmp | RSH | rsh
1 | teralab-x16 | 8 | /remote/users/tmp | RSH | rsh
```

A detailed explanation of the hosts file follows. **This file is typically created and provided by the end user of the application.**

FIELD	VALUE	PURPOSE
1 ('1')	Integer – can be '0' or '1'	A '0' signifies that the host cannot be used, a '1' denotes a usable WORKER machine
2 ('teralab-x')	String, valid hostname	This is the hostname of a valid WORKER machine. This field is only used for RSH and SSH workers. For SH, SGE, LSF, PBS and RTDA, this field is empty
3 ('4')	Integer – any valid integer	This number is used to denote the number of worker slots on this host (RSH, SSH and SH) or farm (LSF, SGE, PBS and RTDA). -1 indicates unlimited.
4 ('/remote/users/tmp')	String – a valid, writable directory	This directory is used for local data storage for an application, if needed
5 ('RSH')	String – command mode	The string signifies the command-mode used to connect to the WORKER machine. The valid entries RSH, SSH, SH, LSF, SGE, PBS and RTDA.
6 ('rsh')	String – the connection command	The given string denotes the remote command used to connect to the WORKER machine.

Worker Information File

A very important aspect of the DP master registration process is supplying Worker Information to the DP System. This can be done either by using a CDPL API or by providing a Worker Information file to the DP system. The Worker Information file provides the following information to the DP system.

- The type of worker required by the master; this is application dependent

- The worker executable name and location
- The worker configuration for a DP run.

A typical Workers Information File has the following format.

```
# Sample WORKER Information File
# worker_type; max_workers; min_workers; parameterized worker_executable invocation

100; 2; 1; worker_exe {other parameters}
```

FIELD	DESCRIPTION
100	The type of WORKER for executing the MASTER tasks
2	The maximum number of WORKERS – CDPL attempts to start as many workers it can, up to the maximum specified in this file
1	The minimum number of WORKERS needed for successful operation – it should be at least ‘1’
worker_exe	The worker executable. The worker executable can be prepended with a wrapper script that can be very useful in providing visibility into worker operation. A sample wrapper script (START_WORKER) is provided in the tutorials’ subdirectory of the CDPL SDK.
Other Parameters	Application specific parameters the worker will need

Host File Configuration Editor

To create a new CDPL host file, click on New->Host File or click on the Host File icon in the tool bar.

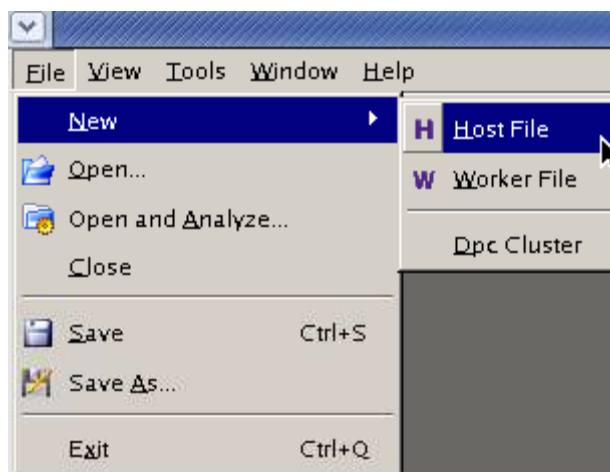


Figure 33



Figure 34

Specify the command mode (mode used to launch DP workers across nodes), temporary directory (optional) and enter the names of the hosts in each row provided by the Configuration Editor Window.

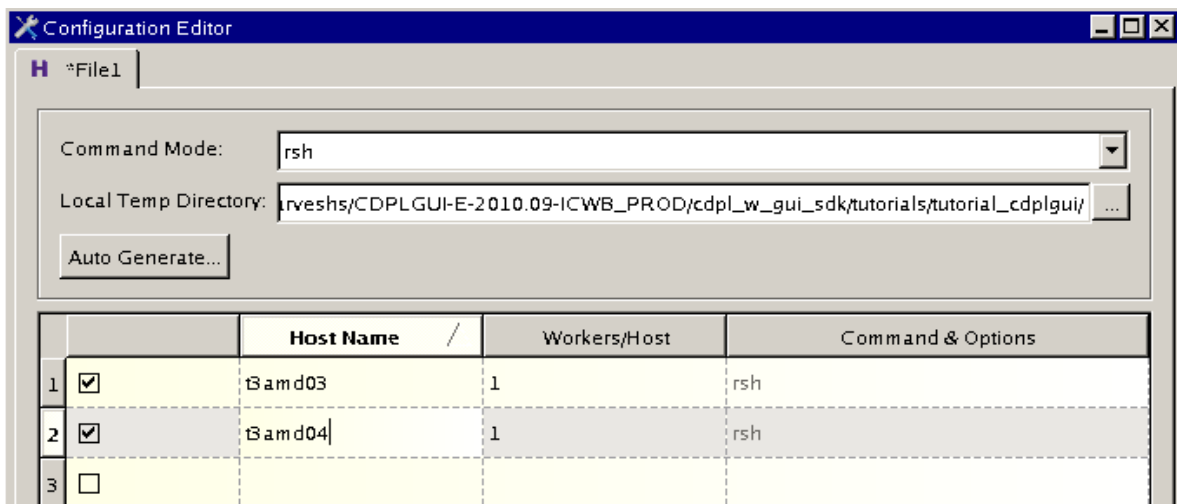


Figure 35

Save the file to an appropriate directory when you are done.

The Host File Configuration Editor has the following features.

- The file name is shown in the tab. Move the cursor to the tab to show the full file name (path + file name) in the tooltip.
- An asterisk is added at the beginning of the file name if some values have been changed.
- Double click in a table cell to start editing in an empty row and all values from the previous row will be copied to this row for user's convenience. For the first row, default values will be used.
- An empty row is always added to the end of the table.
- Right-click the menu items "Add Row" and "Remove Row(s)" to add or remove a row(s).

Some of the things one has to keep in mind while using the host file configuration editor are:

- Host names are only entered in rsh and ssh modes. The Host Name field is disabled from editing in LSF, SGE PBS, RTDA and SH modes.

- Host names are validated in rsh and ssh modes. The Configuration Editor runs the “ping” command to check if the host is available or not. If the host name is empty, or the host is not available, or there is a duplicate definition for this host, the table cell’s background is set to red.
- The local temp directory should be writable on the host systems.
- “Command & Options” is only editable in LSF, SGE, PBS and RTDA modes. It is disabled from editing in RSH, SSH and SH modes.
- In this release, there is no validation for “Command & Options”.

The “Auto generate...” button is only enabled for host configuration files. The auto generation dialog is used to create multiple definitions for hosts with similar names. For example, to create host definitions for hosts from “teralab-10” to “teralab-20” with command mode “rsh”, we have the following input values.

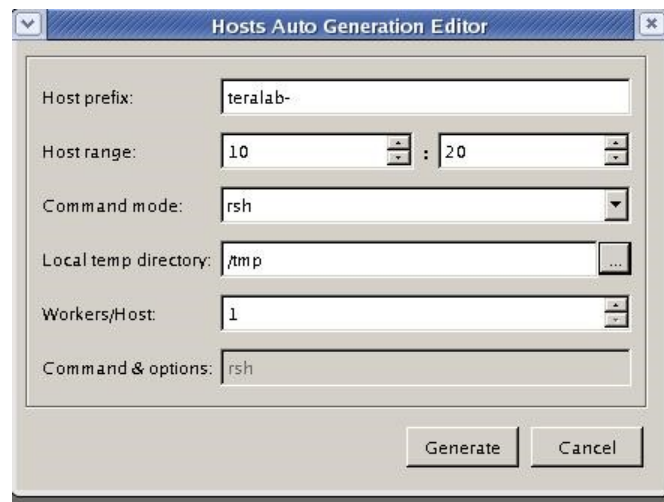


Figure 36

Press the “Generate” button to get host definitions as shown below.

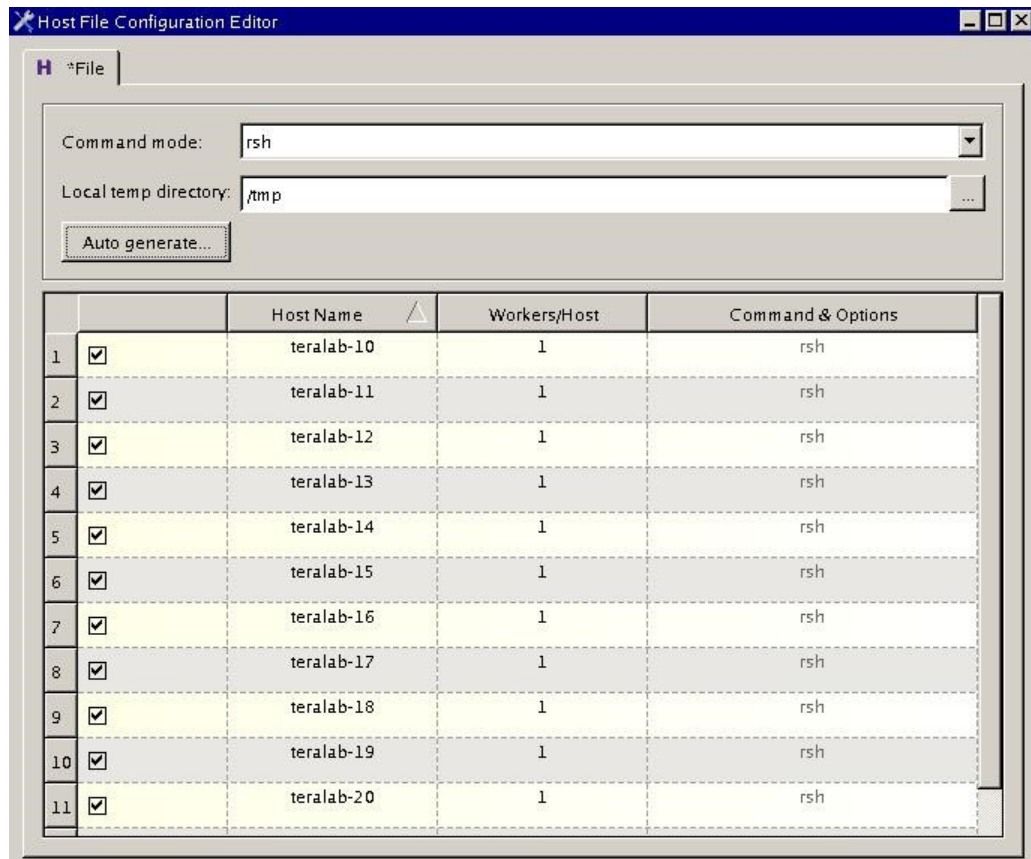


Figure 37

Worker File Configuration Editor

To create a new DP workers file, click on New->Worker File or click on the Worker File icon in the tool bar. Enter the Worker Type, minimum number of workers, maximum number of workers, select the Worker Executable and specify its input arguments (if any) in the last column.

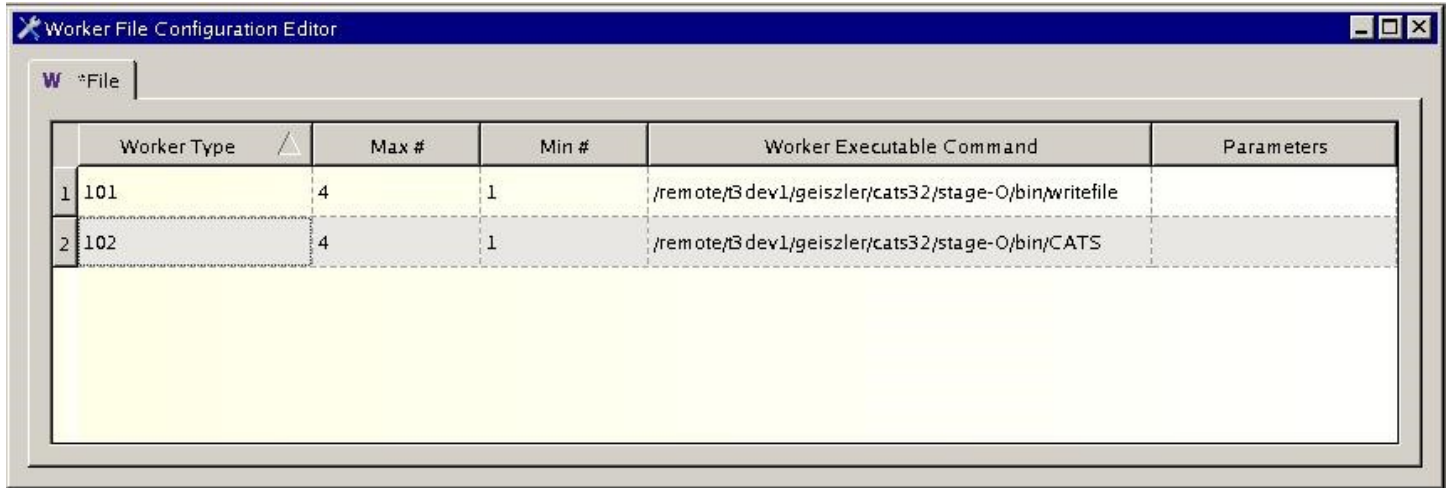


Figure 38

Some of the things one has to keep in mind while configuring a worker information file are:

- For each row, the “Max #” value cannot be smaller than the “Min #” value.
- “Worker Executable Command” cannot be empty. It must be an executable file name.
- For this release, no validity check is done on the “Parameters” value.

Some scenarios where you may want to edit a worker information file are the following.

- Add a new worker type
- Change the number of minimum and maximum workers
- Add new parameters to the worker environment.
- Change the worker type of a particular worker.

Cluster Manager

The Cluster Manager provides a user interface for managing sets of hosts to be used for distributed processing. Each set of hosts is called a cluster. Each cluster has a corresponding DP hosts file.

Creating a Cluster

Go to View->Cluster Manager to bring up the Cluster Manager window.

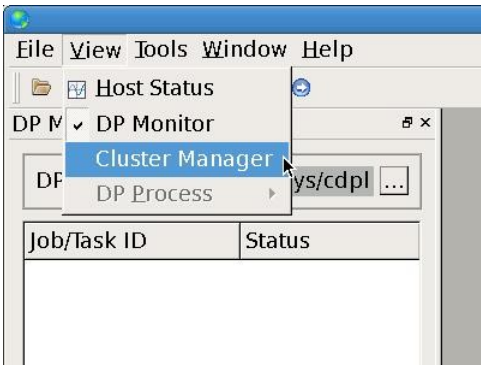


Figure 39

Select “DP Clusters” and right-click “New”. A new cluster with an arbitrary name will appear. You can select and edit the cluster name and description to enter your own information about the cluster.

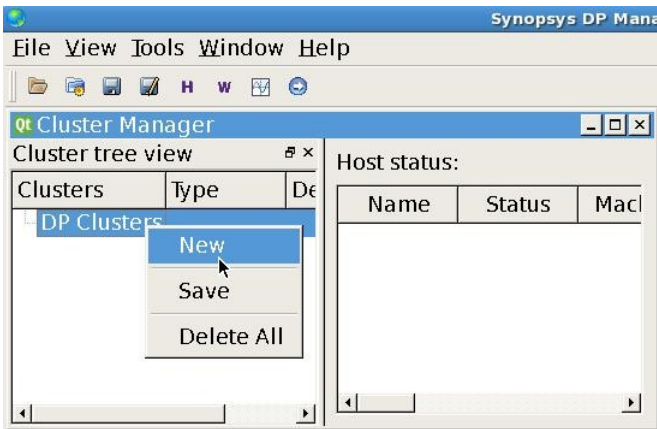


Figure 40

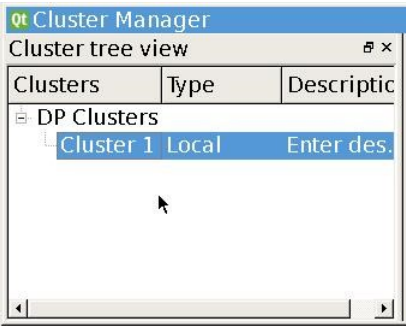


Figure 41

Adding Hosts

Select the cluster in the Cluster Tree View under “DP Clusters”. Right-click and select “Setup”. A Host File Configuration Editor window comes up. Enter the host information for your cluster and save the hosts file.

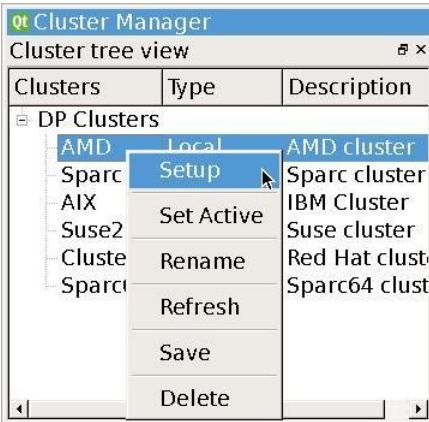


Figure 42

Cluster Host Status

The user interface is divided into two views – the Cluster Tree view and Host Status view. To view the status of the hosts in a cluster, select the cluster in the tree view under “DP Clusters”. The first time you select the cluster; the Cluster Manager will retrieve the status information of each host in the cluster and display it in the Host Status pane. This information will be cached, so that on later selections of the same cluster, the same information will be displayed. You can retrieve and display the updated status of the hosts in the cluster by selecting the cluster and right-clicking “Refresh”.

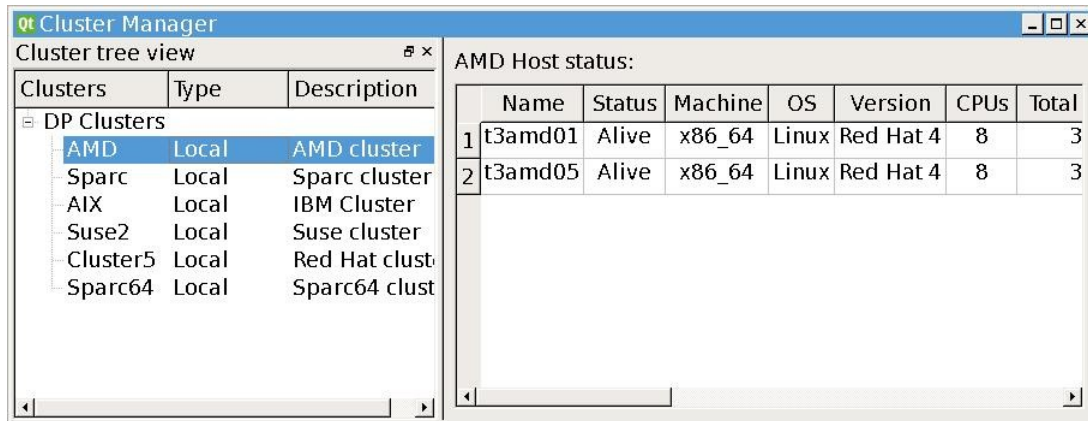


Figure 43

Help Assistant

The DP Manager provides a help module to view its documentation. To access it, go to the Help menu. Select “DP Manager User Guide” or “DP Manager FAQ”. These will invoke “Acrobat Reader” and display the appropriate manual in a new window.

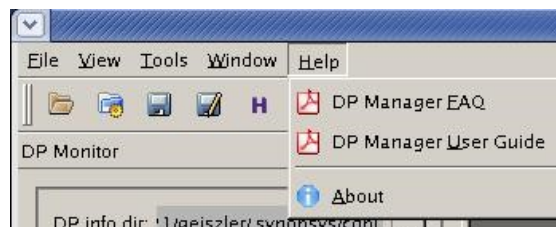


Figure 44

You will need Acrobat Reader version 5.0.5 or greater to be installed on your system and be present in your path variable.

Operational Environment

The following environment variables are available and may be used for operation of the DP Manager and CDPL.

Required Variables

- DISPLAY
- CDPL_HOME
 - Used to locate documentation and relevant DP binaries and scripts. It should point to the root of the CDPL installation.

Optional variables

- CDPL_BCASTDIR
 - Set a location for the DP broadcast files. When unset, the default location \$HOME/.synopsys/cdpl is used
- CDPL_LOGDIR
 - The location of the DP log files.

When starting a DP run from DP Manager, the DP run environment will inherit these environment variables from DP Manager. When starting a DP run outside DP Manager, you must make sure that these environment settings are the same in both DP Manager and the DP run environment.

Follow the given steps to run dpmanager.

- CDPL_HOME should be set to a valid location.
- dpmanager should be in your search path.
- Invoke the DP Manager as follows:

```
% dpmanager &
```

Common Usage Scenarios

This section describes some of the common scenarios that the DP Manager is designed for.

1. Initiate a DP Run

In this scenario the DP Manager is used to start a new DP Run.

- 1) Set appropriate DP environment variables.
- 2) Start the GUI.

- 3) Modify DP configuration files such as the hosts or workers file using the configuration editor. Add/remove hosts from the hosts file, change the number of workers for a particular worker type in the workers file.
- 4) Use the Tools->Start DP Run menu item to start the DP run.
- 5) Select the application master script or binary in the File Chooser dialog.
- 6) Start the DP run by pressing the Open button.
- 7) While the DP run is executing, use the GUI to monitor the progress.
- 8) View the master application output through the View->DP Process menu.

2. Connect to an existing DP run

In this case the user starts a DP run through the command line and then connects the GUI to this run.

- 1) Start a DP run through the command line.
- 2) Start the GUI
- 3) The GUI will now connect to the master by locating its broadcast file in `$CDPL_BCASTDIR`. If `CDPL_BCASTDIR` is not set, the master will search for the broadcast file in `$HOME/.synopsys/cdpl`.

Important: Both the GUI and the CDPL master application must have the same setting for the broadcast directory.

- 4) While the GUI is connected to the master, use the GUI to monitor the progress of the job.

3. Use the GUI as a monitoring tool

- 1) Either initiate a new CDPL run as shown in the first scenario or connect to an existing run as shown in the second usage scenario.
- 2) Observe Job and Task status information in the DP Monitor. This information is presented as a hierarchical tree.
- 3) Click on the Master name at the topmost level of the tree to reveal the job.
- 4) Click on the Job to expand its list of tasks. One can see the status of each task and the name of the worker to which the task has been assigned.
- 5) Select the menu item View->Host Status to observe the CPU and Memory status of the worker systems that are executing the job tasks.
- 6) In the monitor tree view, select the master or job to view the aggregate system status of all the workers. Select a task or worker to view the system status of a specific worker.
- 7) You can save a snapshot of the CPU or Memory graph by right-clicking on the graph.

4. View log information about a job

Use the DP Monitor to view information about the job

- 1) Once the Job is completed, use the DP Analyzer to analyze the master log file. Select the Job in the tree view and right-click. Then select "Analyze Log File". The DP Analyzer collates worker and

task statistics that can be viewed in a Gantt chart, a Treemap view or a Summary. The Summary will show an “efficiency” measure for the DP run.

- 2) View the master log file by selecting the master, job or a task in the tree view. Right-click and select “Open CDPL Master Log”. The master log file will be displayed.
- 3) View a worker log file by selecting a worker in the tree view. Right-click on the worker and select “Open CDPL Worker Log”. The worker log file will be displayed.

5. Kill a DP Run and start again

- 1) DP Manager provides users the ability to kill a DP run if something goes wrong. The processes are not blindly killed and are given a chance to clean up.
- 2) Start a DP run from the GUI or from the command line.
- 3) Once the job is visible in the DP Monitor, kill the DP run by right-clicking on the job in the tree view and selecting “Kill Job”.
- 4) The job and task status will show that it has been killed. The job will have status “Killed”. Any tasks not completed will have status “Interrupted”. The status may not change immediately because the application may take some time to clean up and shut down.

6. Create and edit a new Hosts file

This scenario explores the use of the Host File Configuration Editor to create and edit a new hosts file.

- 1) Click on File->New->Host File menu item. The new host file configuration editor window is displayed.
- 2) Select the command mode from the list of supported modes.
- 3) Select the local temporary directory or leave blank.
- 4) Fill in the fields provided by the editor. For command modes “rsh” and “ssh, the Host Name field is available, but the Command and Options field is fixed. For command modes “SGE” and “LSF”, the opposite is true. In these modes, the Host Name field is not available, but the Command and Options field is available.
- 5) Use Save or Save As from the File menu or toolbar to save the new hosts file.
- 6) The file may be edited later by using the File->Open menu.

7. Create and edit a new workers file

This scenario illustrates the use of the Worker Configuration editor to create and edit a workers file.

- 1) Click on File->New->Worker File menu item. The new Worker File Configuration editor window is displayed.
- 2) Fill in the fields provided by the editor.
- 3) Use Save or Save As from the File menu or toolbar to save the new hosts file.
- 4) The file may be edited later by using the File->Open menu.

8. Create and edit a DP cluster

This scenario illustrates the use of the Cluster Manager to create and edit a DP cluster. A DP cluster is a set of hosts that will be used for a DP run. Each cluster has a corresponding CDPL hosts file. The user interface is divided into two views – the Cluster Tree view and Host Status view.

- 1) Go to View->Cluster Manager to bring up the Cluster Manager window. Select DP Clusters and right-click New. A new cluster with an arbitrary name will appear. You can select and edit the cluster name and description to enter your own information about the cluster.
- 2) To add hosts to the cluster, select the cluster in the Cluster Tree View under DP Clusters. Right-click and select Setup. A Host File Configuration editor window will come up. Enter the host information for your cluster and save the hosts file.
- 3) You may add additional clusters by following steps 1) and 2) again.
- 4) Save the cluster information to a file using Save or Save As.
- 5) To view the status of the hosts in the cluster, select the cluster in the tree view under DP Clusters. The first time you select the cluster, the Cluster Manager will retrieve the status information of each host in the cluster and display it in the Host Status pane. This information will be cached, so that on later selections of the same cluster, the same information will be displayed. You can retrieve and display the current status of the hosts in the cluster by selecting the cluster and right-clicking Refresh.